

WORLD-MODEL TRANSFER FOR DATA-EFFICIENT TIME-SERIES FORECASTING

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ABSTRACT

Rendering a time series as an image and forecasting it by inpainting is the recipe behind the strongest published zero-shot forecasters, all of which inpaint with an ImageNet autoencoder. A video backbone, pretrained on how signals evolve rather than on how they look, ought to be the better prior. We find that it is, but only once the transfer objective is changed. Continuing VideoMAE’s masked-pixel pretraining on a corpus of rendered series improves its own held-out loss monotonically while making forecasts monotonically worse over a $35\times$ range of training budget. The two never meet, because the read-out that converts pixels back into numbers sits outside the loss. We put it inside: the backbone is trained as a world model over rendered series, observing past frames, predicting future ones, and scored by a differentiable read-out that decodes those frames into standardised values. The budget curve inverts. With period-aligned rendering, frame-scale augmentation and 32-frame clips, a single checkpoint under a single inference rule attains 0.285 mean MSE across the six standard long-sequence benchmarks, ahead of VisionTS++ (0.289) and VisionTS (0.315), using 1/13 of VisionTS++’s continual-pretraining steps and 0.3% of its data. Swapping only the initialisation, between image, video and none at all, shows that what transfers is a prior over dynamics, not capacity and not any particular video objective.

1 INTRODUCTION

Zero-shot forecasters are trained once and applied to datasets they have never seen. The models that do this well today — Moirai (Woo et al., 2024), Chronos (Ansari et al., 2024), Time-MoE (Shi et al., 2025), TimesFM (Das et al., 2024) — get there by pretraining on time-series corpora measured in tens or hundreds of billions of observations. That is a reasonable way to spend resources if one has them, and a barrier if one does not. The question we take up is how far a competitive zero-shot forecaster can be pushed when the time-series pretraining budget is small, by starting from a model that has already learned something about how signals evolve.

One answer already exists in the visual direction. VisionTS (Chen et al., 2025) renders a series as an image and lets a frozen ImageNet masked autoencoder (He et al., 2022) inpaint the missing right-hand portion; VisionTS++ (Shen et al., 2025b) continues that backbone’s pretraining on rendered series and reports the strongest published zero-shot numbers on the standard long-sequence protocol. Both inherit a prior over spatial structure from a model that has never seen anything change over time. Video pretraining offers a different inheritance: a representation built from sequences, and, in the predictive variants, an explicit forward model (Tong et al., 2022; Assran et al., 2025). Whether that buys anything for forecasting is an empirical question, and the useful form of it is about efficiency rather than ceiling: *does a video-pretrained backbone reach a given forecasting accuracy under a smaller time-series adaptation budget?*

Answering it is not a matter of swapping the backbone. A numeric series has to become a video whose axes mean something; masked reconstruction recovers a token from neighbours on both sides rather than predicting a future from a past; and the error the model minimises is measured in gray levels while the task is scored in the units of the series. These gaps are not cosmetic: continuing VideoMAE’s native objective on a 654M-observation corpus of rendered series drives its held-out pixel loss down monotonically and its forecast error up monotonically across the checkpoints we evaluated (Section 4.6).

We therefore treat the video backbone as an initialisation for a forecasting model and build the training stage around the forecasting task. Three design choices carry the framework (Figure 1). **Period-aligned rendering** lays one period across the horizontal axis of a frame and value on the vertical axis, so that consecutive frames express evolution from one period to the next rather than continuation of a line. **Forecast-aligned prediction and supervision** masks a trailing block of frames, predicts it from the visible past only, and decodes the predicted frames back into values with a differentiable read-out, so the training loss is computed on numbers rather than on pixels. **Temporal-scale alignment** samples during training the same periods-per-frame scales the inference rule will use, so the model is trained on the resolutions it is asked to work at. Together they make the inherited weights usable: the budget curve that ran backwards under the native objective runs forwards under this one.

Under this framework, 60k optimizer updates at an effective batch of 32 on 654M observations — a corpus roughly 0.3% the size of the LOTSA archive VisionTS++ trains on, and roughly 1/27 of its sample presentations (Section 4.3) — yield one checkpoint that, under one inference rule, reaches a mean MSE of 0.285 over the six standard long-sequence datasets and four horizons. That is the lowest mean MSE among the models we compare on this protocol, ahead of VisionTS++ large (0.289) and base (0.291); mean MAE is 0.328 against their 0.326, which is worse and we report it as such. Changing only the initialisation shows where the video prior contributes: at a 5k-step budget both video initialisations beat an ImageNet initialisation by 3–5% at every horizon and random initialisation is the worst arm by a wide margin, while by 20k steps the image initialisation has caught up — which bounds the claim to the low-budget regime (Section 4.4).

Contributions.

- A transfer framework that turns a video-pretrained backbone into a zero-shot time-series forecaster under a small time-series pretraining budget: period-aligned video rendering, future-block prediction supervised through a differentiable numeric read-out, and a training scale distribution matched to the inference rule (Section 3).
- Zero-shot accuracy competitive with the strongest published visual and time-series foundation models on the six-dataset long-sequence protocol, from 60k updates on 654M observations, with the budget reported in corpus size, optimizer updates and sample presentations rather than as a compute claim (Sections 4.2 and 4.3).
- Controlled evidence for where the gain comes from and where it stops: an initialisation study at matched size and budget with a random-init control, an ablation of the forecast-aligned objective, and the diagnosis of why naive continual pretraining fails (Sections 4.4–4.6).

2 RELATED WORK

Time-series foundation models. Moirai (Woo et al., 2024), Chronos (Ansari et al., 2024), Time-MoE (Shi et al., 2025), Timer (Liu et al., 2024), TimesFM (Das et al., 2024) and MOMENT (Goswami et al., 2024) pretrain on large time-series corpora with tokenisation, quantisation or patching designed for the modality, and are evaluated zero-shot on held-out datasets. They set the accuracy we compare against, and they also set the resource expectation we are trying to undercut: the corpora are measured in tens or hundreds of billions of observations.

Forecasting with visual backbones. VisionTS (Chen et al., 2025) renders a univariate series into a periodicity-aligned image, masks the right-hand portion, and inpaints it with a frozen ImageNet masked autoencoder (He et al., 2022); decoding the completion gives a forecast that is competitive zero-shot without any time-series pretraining. VisionTS++ (Shen et al., 2025b) keeps the image construction and continues pretraining the backbone on LOTSA (Woo et al., 2024), with colourised multivariate rendering and a multi-quantile head, and reports the strongest zero-shot numbers on this protocol.

Two points of contrast matter for what follows. First, the visual prior that survives this transfer appears to be small: SVTime (Shen et al., 2025a) matches VisionTS with a 215k-parameter model carrying three inductive biases, and OccamVTS (Lyu et al., 2026) distils a vision forecaster into

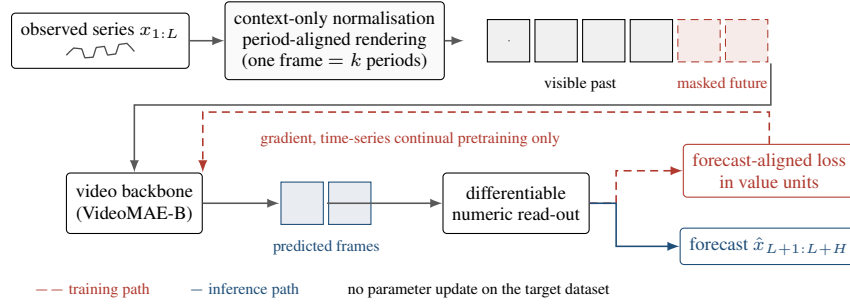


Figure 1: The transfer framework. A series is normalised with the statistics of its visible context only and rendered so that one frame holds k periods, which puts value on the vertical axis and phase on the horizontal one; the trailing frames are masked. A video backbone predicts those frames, and a differentiable read-out decodes them back into values. During time-series continual pretraining the loss is taken on the decoded values (Section 3.3); at test time the same read-out emits the forecast and no parameter is updated on the target dataset. The backbone enters this pipeline with weights from video pretraining, which is the only stage that sees video.

1% of its parameters. Second, and contrary to a natural reading of our own results, VisionTS++ does not optimise pixel reconstruction alone: it trains against a multi-quantile (pinball) objective on the forecast. The objective mismatch we report in Section 4.6 is therefore a statement about the configuration we tested — VideoMAE’s native masked-pixel objective continued on rendered series — and not a criticism of the image-based systems we compare with.

Video pretraining and future prediction. VideoMAE (Tong et al., 2022) reconstructs masked spatiotemporal tubes; the temporal structure it acquires is implicit in the representation rather than expressed as a forward model. V-JEPA (Bardes et al., 2024) and V-JEPA 2 (Assran et al., 2025) predict the latent representation of masked regions from context and are used downstream for planning, which is closer to the world-model formulation in the sense of LeCun (2022) and Hafner et al. (2025). We use both families as initialisations. Our training stage is world-model-style in one specific sense — the model maps an observed past to a predicted future and is scored on that prediction — and we make no claim about action conditioning, planning or physical reasoning, none of which we test.

Transfer regularisation. Continual pretraining of a strong backbone on a narrow corpus can erase the prior that motivated choosing it. L2-SP (Li et al., 2018) decays weights toward their initial values rather than toward zero. It is an existing method, we use it only as an ablation arm, and we report that it helps at short clip lengths and is inert in the configuration of our main model.

3 A TRANSFER FRAMEWORK FROM VIDEO PRETRAINING TO FORECASTING

Overview. The framework is one pipeline, run in two directions (Figure 1):

observed series $\rightarrow$ context-only normalisation $\rightarrow$ period-aligned frames $\rightarrow$ visible past / masked future $\rightarrow$
video backbone $\rightarrow$ predicted frames $\rightarrow$ numeric read-out $\rightarrow$ {forecasting loss, forecast}.

During time-series continual pretraining the last step produces a loss; at test time it produces the forecast. The design question is what has to be true of the middle of that pipeline for weights learned on natural video to be worth anything at the end of it, and the three subsections below give our three answers: a representation in which frames carry temporal evolution (§3.2), a training signal that is a forecast rather than a reconstruction (§3.3), and a match between the temporal scales seen in training and those used at inference (§3.4).

3.1 THREE STAGES, AND WHAT ZERO-SHOT REFERS TO

(i) **Video pretraining** is inherited: a public VideoMAE-B checkpoint trained on Kinetics-400 (Tong et al., 2022), and in the initialisation study V-JEPA checkpoints (Assran et al., 2025). (ii) **Time-series continual pretraining** is the stage this paper designs, on a corpus containing none of the evaluation datasets (§3.6). (iii) **Zero-shot evaluation** applies the resulting checkpoint with no parameter update and no per-dataset tuning. Design decisions and checkpoint selection used held-out pretraining data only.

3.2 A. PERIOD-ALIGNED VIDEO REPRESENTATION

A series is turned into a video, not into an image. Let P be the dominant period, taken from the sampling frequency as in Chen et al. (2025), and let k be the number of periods packed into one frame. A clip of F frames covers FkP observations: frame f holds the observations of periods $[fk, (f+1)k)$, drawn as an area chart in a 224×224 frame. Within a frame the horizontal axis is phase inside the period and the vertical axis is value; *across* frames the axis is time measured in periods. Consecutive frames therefore differ by whatever changed from one period to the next — level, amplitude, shape — and a seasonal series renders as an almost static video while a drifting or regime-switching one renders as a moving one.

This is what separates the representation from image folding. Folding a series into a 2-D image also aligns periods, but the model then sees one static picture to complete. Here the period axis is the model’s *time* axis — the axis a video backbone was pretrained to extrapolate along — and the masked region is a block of future frames.

Concretely, values are standardised with the mean and standard deviation of the visible context only, clipped to $z \in [-3, 3]$, and mapped to a column height

$$h = 0.5 + 0.4 \frac{z}{3} \in [0.1, 0.9], \quad (1)$$

with the column filled below the boundary row $\text{round}((1-h)(\text{IMG}-1))$ and background above it. The drawing constants and the grid are given in Appendix A. The main model uses $F = 32$ frames with a tubelet size of 2, giving 16 temporal positions; F is a context-length setting rather than a contribution, and we report what changes with it in Appendix C.

3.3 B. FORECAST-ALIGNED PREDICTION AND SUPERVISION

Predict a future block, not a random subset. The trailing h_p frames of the clip are masked and must be predicted from the visible past. This is the part of the formulation we call world-model-style: the model maps an observed past to a predicted future state, in the sense of LeCun (2022), and unlike the backbone’s native masked-reconstruction objective it never sees a neighbour on the far side of what it is predicting. With probability 0.3 the leading tubelets are additionally hidden so that shorter contexts are also trained on, which is the regime the shorter evaluation datasets present at test time. A fraction of batches retains the backbone’s original tube masking, as a regulariser; Appendix A gives the exact schedule and which loss applies to which batch type.

Supervise in value space. A predicted frame is a picture, and the quantity the task is scored on is a number, so the decoder that converts one into the other belongs inside the loss. For a predicted frame $\hat{x}$ we compute a differentiable soft occupancy per pixel and sum it down each column,

$$s(\hat{x}_{r,c}) = \text{clamp}\left(\frac{g_{\text{grid}} - \hat{x}_{r,c}}{g_{\text{grid}} - g_{\text{fill}}}, 0, 1\right), \quad \hat{h}_c = 1 - \frac{\text{IMG} - \sum_r s(\hat{x}_{r,c})}{\text{IMG} - 1}, \quad (2)$$

which inverts the drawing rule of Equation 1 up to rounding: the thresholds are chosen so that background and grid pixels contribute zero and filled pixels contribute one, and partially filled boundary pixels contribute fractionally. Inverting Equation 1 gives $z = (h - 0.5) \cdot 3/0.4$, and the training objective is

$$\mathcal{L} = \text{MSE}(\hat{x}, x) + \lambda \left(\frac{3}{0.4}\right)^2 \text{MSE}(\hat{h}_c, h_c), \quad \lambda = 0.1. \quad (3)$$

The constant converts height units into the units of the context-standardised series, so the second term is a squared error on values rather than on gray levels. It is computed column-wise and is

therefore independent of P and of how many periods a frame holds. The pixel term is retained because early in training the read-out of a blurry prediction carries little signal.

One clarification about units: this error is in the *context-instance* z-scale, since each window is standardised by its own visible statistics, while the benchmark reports errors in a scale fitted on the target dataset’s training split. The two differ by the ratio of those standard deviations, which varies per window. The objective is therefore an error on values rather than on pixels, which is the property we need, but it is not numerically identical to the benchmark metric.

3.4 C. TEMPORAL-SCALE ALIGNMENT

A forecaster must cover horizons from one period to many, and in this representation the horizon a single predicted block can cover is $h_p k P$. Inference therefore varies k with the requested horizon (§3.5), and training must cover the same range or the model meets resolutions it never saw. We sample $k \in \{1, 2, 4\}$ during training with probabilities 0.5/0.3/0.2, which is the distribution the inference rule draws from. This is a training–inference consistency property of the framework rather than a generic augmentation: the set of scales is determined by the inference rule, not chosen for diversity.

3.5 INFERENCE

Given the period P and the requested horizon H , the rule selects the number of periods per frame and the number of forward passes, with no per-dataset tuning and no checkpoint selection. Short horizons are covered by a single predicted block whose outputs are averaged over the admissible scales; long horizons increase k and, when one block still cannot span H , roll the prediction forward autoregressively. The resulting forecast is truncated to H . Multi-scale averaging and rollout both cost additional forward passes, so the system is single-*checkpoint* and single-*rule* but not single-pass; Appendix D gives the configuration and the pass count for each of the 24 evaluation cells.

3.6 PRETRAINING CORPUS

The corpus is built from LOTSA (Woo et al., 2024): 98 subsets, 184,013 series, 654.2M observations, with synthetic series mixed in during sampling and subsets drawn in proportion to the square root of their size. None of ETTh1, ETTh2, ETTm1, ETTm2, electricity or weather is included. The corpus does contain Monash `traffic_hourly` (Godahewa et al., 2021), which is the same data as the LSF traffic benchmark, so traffic is excluded from our evaluation rather than reported as a zero-shot result. Appendix A lists the subset composition and the synthetic-mixing rule.

4 EXPERIMENTS

The experiments answer four questions in order: what accuracy the framework reaches under a small time-series pretraining budget (§4.2, §4.3); what the video initialisation contributes and where that contribution stops (§4.4); what the transfer design contributes (§4.5); and why the naive alternative fails (§4.6).

4.1 SETUP

Protocol. We follow the zero-shot long-sequence protocol of Chen et al. (2025): Time-Series-Library splits (Wu et al., 2023) (ETT 12/4/4 months, others 70/10/20), a `StandardScaler` fitted on the target dataset’s training portion, all channels, stride-1 test origins, $H \in \{96, 192, 336, 720\}$, MSE/MAE on standardised values, and their per-dataset context lengths and periods (Appendix A). Running their released checkpoint through our harness reproduces their published numbers to three decimals (ETTh1 at $H=96$: 0.3527/0.3833 against 0.353/0.383). Zero-shot here means no parameter is updated on any evaluation dataset and no evaluation dataset appears in the pretraining corpus; the model is trained once and applied unchanged, with one checkpoint and one inference rule for all 24 cells.

Table 1: Zero-shot long-sequence forecasting. Each entry averages the four horizons $H \in \{96, 192, 336, 720\}$; MSE and MAE are computed on standardised values. Our model uses one checkpoint and one inference rule for all 24 cells. Baselines are quoted from Shen et al. (2025b) and Chen et al. (2025); ‘–’ means the source does not report that dataset. **Bold**: best. Underlined: second best. The last column counts first places over the twelve dataset columns.

Model	ETTm1		ETTm2		ETTh1		ETTh2		ECL		Weather		Avg		1 st
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
Ours	0.352	0.369	0.241	0.302	0.404	0.416	0.335	0.376	0.163	0.254	0.215	0.252	0.285	0.328	6
VisionTS++ (ViT-L)	0.354	0.369	0.244	0.298	0.403	0.418	0.327	0.365	0.181	0.264	0.226	<u>0.243</u>	<u>0.289</u>	0.326	4
VisionTS++ (ViT-B)	0.360	<u>0.372</u>	<u>0.244</u>	0.298	0.402	<u>0.416</u>	<u>0.333</u>	<u>0.370</u>	0.184	0.265	<u>0.222</u>	0.241	0.291	<u>0.327</u>	2
VisionTS	0.374	<u>0.372</u>	0.318	0.366	0.390	0.414	<u>0.333</u>	0.375	0.207	0.294	0.269	0.292	0.315	0.352	2
Moirai _S	0.448	0.410	0.272	0.321	0.400	0.424	0.341	0.379	0.233	0.320	0.242	0.267	0.323	0.353	–
Moirai _B	0.382	0.388	0.276	0.320	0.434	0.439	0.346	0.382	0.188	0.274	0.238	0.261	0.311	0.344	–
Moirai _L	0.390	0.389	0.317	0.366	0.510	0.469	0.354	0.377	0.188	0.273	0.260	0.275	0.337	0.358	–
Chronos _S	0.640	0.500	0.310	0.350	0.545	0.472	0.424	0.430	0.220	0.284	0.300	0.318	0.407	0.392	–
Chronos _B	0.646	0.500	0.295	0.338	0.591	0.468	0.406	0.411	0.215	0.279	0.293	0.315	0.408	0.385	–
Chronos _L	0.556	0.465	0.300	0.341	0.589	0.466	0.455	0.427	0.204	0.274	0.279	0.306	0.397	0.380	–
Time-MoE _S	0.394	0.416	0.316	0.361	0.400	0.424	0.367	0.404	–	–	0.266	0.297	–	–	–
Time-MoE _B	0.376	0.406	0.349	0.380	<u>0.394</u>	0.420	0.405	0.415	–	–	0.270	0.300	–	–	–
Timer	0.487	0.457	0.328	0.347	0.444	0.457	0.358	0.407	–	–	0.304	0.331	–	–	–
TimesFM	0.433	0.419	–	–	0.473	0.444	0.392	0.406	–	–	–	–	–	–	–
MOMENT	0.670	0.537	0.316	0.371	0.684	0.566	0.362	0.410	0.765	0.687	0.294	0.326	0.515	0.483	–

Ablations and baselines. Ablations and initialisation studies use five datasets (ETT×4 and weather) at three horizons, which is what one evaluation budget covers; the six-dataset four-horizon grid is reserved for the main table. Five- and six-dataset means are never mixed and every table says which it reports. Baseline numbers are quoted from Shen et al. (2025b) and Chen et al. (2025); we did not retrain them.

4.2 ACCURACY UNDER A SMALL CONTINUAL-PRETRAINING BUDGET

Table 1 reports four-horizon averages. Our model attains the lowest mean MSE among the models compared on this protocol, 0.285, ahead of VisionTS++ large (0.289), and takes the most first places over the twelve dataset columns. The gains are concentrated in electricity (0.163 against 0.181) and weather (0.215 against 0.222). Two datasets go the other way: ETTh1 to VisionTS (0.390 against our 0.404) and ETTh2 to VisionTS++ large (0.327 against 0.335), both hourly sets where a longer tuned context helps at long horizons. Mean MAE is 0.328 against VisionTS++ large’s 0.326: we are behind on that metric, and we have no uncertainty estimate that would let us call the difference immaterial.

At 20k updates instead of 60k the same recipe reaches 0.288/0.331, still first on mean MSE here at a third of the training.

Averaged over the six datasets, error grows from 0.223/0.281 at $H=96$ to 0.349/0.377 at $H=720$; the full per-horizon breakdown is Table 5 in Appendix B, and the two datasets we lose overall are lost at the mid-to-long horizons rather than uniformly.

4.3 PRETRAINING BUDGET

Table 2 states the resources on both sides in the units each paper reports. We use a corpus of 654.2M observations and 60k optimizer updates at an effective batch of 32. We report corpus size, optimizer updates and sample presentations separately, and we do not convert them into a compute claim: the models differ in size, the number of time points and visual tokens per sample differs between the two renderings, and we do not know the baselines’ hardware or wall-clock. The comparison also excludes the original video pretraining of our backbone, which we inherit rather than pay for, exactly as the image-based systems inherit ImageNet pretraining.

Table 2: Time-series pretraining budget, in the units each paper reports. Sample presentations are optimizer updates $\times$ effective batch, i.e. how many rendered windows the model was shown, not a compute measure: the two systems differ in model size and in how many time points and visual tokens a single sample contains. The original video or image pretraining of each backbone is inherited by both and excluded. VisionTS++ figures are as reported in Shen et al. (2025b); ‘not reported’ means the source does not state it. Our corpus is $\approx 0.3\%$ of the LOTSA archive and our main run makes $\approx 1/27$ of VisionTS++’s sample presentations (1/80 for the 20k run).

	Ours (main)	Ours (20k)	VisionTS++
Time-series corpus (observations)	654.2M	654.2M	LOTSa archive, 231B
Optimizer updates	60,000	20,000	100,000
Effective batch (windows/update)	32	32	512
Sample presentations	1.92M	0.64M	51.2M
Backbone	ViT-B (87M)	ViT-B (87M)	ViT-B / ViT-L
Measured wall clock	≈ 20 h, $1 \times A40$	6.3 h, $1 \times A40$	not reported

Table 3: Initialisation study. Rendering, objective, corpus, schedule and inference rule are held fixed; only the weights the model starts from change. Five-dataset mean MSE (ETT $\times 4$ and weather) at $H=96/336/720$. Two seeds are listed where both were run. At a 5k-update budget both video initialisations lead the image initialisation; by 20k the image initialisation has caught up, which is why we claim adaptation efficiency rather than a higher ceiling. The V-JEPA arms are size-matched, not architecturally identical (see text).

Initialisation	Pretraining signal	Params	5k updates	20k updates
ImageNet MAE-B	images	87M	0.264 / 0.350 / 0.395	0.244 / 0.329 / 0.373 0.245 / 0.330 / 0.373
VideoMAE-B	video, reconstructive	87M	0.250 / 0.337 / 0.384	0.245 / 0.330 / 0.379
V-JEPA 2.1-B	video, predictive	87M + 23M	0.252 / 0.335 / 0.382	not run
V-JEPA 2-L	video, predictive	300M	0.245 / 0.329 / 0.379	not run
V-JEPA 2-L, random init	none	300M	0.280 / 0.369 / 0.418	not run

4.4 WHAT THE VIDEO INITIALISATION CONTRIBUTES

Table 3 holds the rendering, objective, corpus, schedule and inference rule fixed and changes only the weights the model starts from. At a 5k-step adaptation budget, both video initialisations lead the ImageNet initialisation by 3–5% at every horizon, and they are within 0.002 of each other, so the reconstructive and predictive video objectives are not distinguishable here. The random-init control is the informative row: the same ViT-L architecture, same data, same schedule, reaches 0.280 at $H=96$ against 0.245 for its pretrained twin, and is worse than an image-pretrained backbone $3.5\times$ smaller. Whatever transfers is not capacity.

Two qualifications belong with this table. First, the V-JEPA arms are size-matched rather than architecturally identical: they carry an additional predictor, use EMA targets, and their predictor output head is re-initialised because the pretrained head maps into a teacher space our targets do not use. Second, the advantage is budget-dependent. The right-hand column runs the same comparison at 20k updates, where the image initialisation has caught up and is level or slightly ahead at the longest horizon, so what the video prior buys is adaptation efficiency and not a higher ceiling. Note also that the image-initialised arm is an initialisation control, not a reimplement of VisionTS++, which is a different system with a different rendering and a multi-quantile training objective.

4.5 WHAT THE TRANSFER DESIGN CONTRIBUTES

Table 4 varies the training objective and the budget with everything else fixed. Replacing the forecast-aligned value-space term with pixel-only supervision, holding masking, rendering, corpus and schedule constant, costs 5% at $H=96$, 5% at $H=336$ and 8% at $H=720$: in this matched comparison the value-space term accounts for the measured gain, and its effect grows with horizon.

Table 4: Ablation of the transfer design. Five-dataset mean MSE (ETT $\times$ 4 and weather), 32-frame clips, identical corpus, masking, schedule and inference rule; only the listed component changes.

Configuration	$H=96$	$H=336$	$H=720$
Full framework, 20k updates	0.245	0.330	0.379
pixel-only supervision (no value-space term)	0.257	0.347	0.411
no prior anchoring	0.244	0.330	0.378
Full framework, 5k updates	0.250	0.337	0.384
Full framework, 60k updates	0.242	0.329	0.377

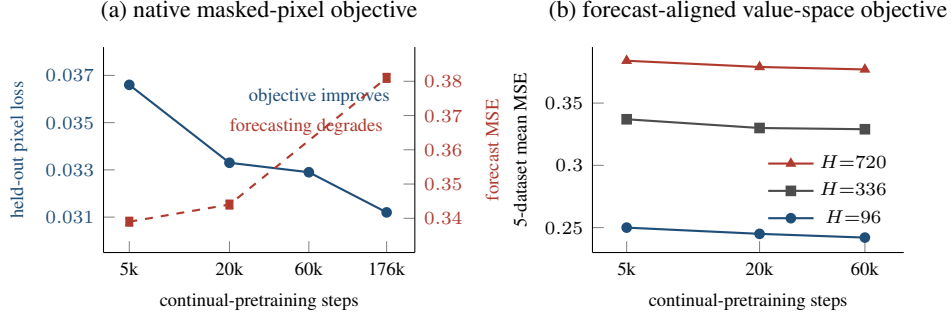


Figure 2: The same backbone and corpus under two objectives. **(a)** VideoMAE’s native masked-pixel objective: its own held-out loss falls monotonically while forecast error rises monotonically over a $35\times$ range of budget. The 60k checkpoint was not evaluated downstream. **(b)** The forecast-aligned value-space objective of Section 3.3: the curve is monotonically improving at every horizon. Panel (a) uses 16-frame clips and panel (b) 32-frame clips; the ablation in Table 4 compares the two objectives at matched clip length.

Prior anchoring (Li et al., 2018) is inert at the 32-frame clip length of the main model, though it matters at 16 frames (Appendix C).

Under this objective the budget curve is monotonically improving: $0.250 \rightarrow 0.245 \rightarrow 0.242$ at $H=96$ for 5k, 20k and 60k updates. Scaling the other two knobs did not help — a larger effective batch is equal or worse at every horizon and a $4\times$ larger in-domain corpus ties ours — Appendix F.

4.6 WHY NAIVE CONTINUAL PRETRAINING FAILS

Figure 2a is the observation that motivated the design. Continuing VideoMAE’s native masked-pixel objective on the rendered corpus improves its held-out pixel loss monotonically across the four checkpoints we evaluated ($0.0366 \rightarrow 0.0312$) while forecast error rises monotonically across the three we evaluated downstream ($0.339 \rightarrow 0.381$). Per cell, the 176k-step model is 5–60% worse than the 20k-step model, with the damage concentrated at long horizons; ETT $\times$ 2 at $H=720$ moves from 0.429 to 0.617. We did not evaluate the 60k native-objective checkpoint downstream, so that point is missing rather than interpolated, and we describe this as a trend over the checkpoints we measured rather than as a scaling law.

Alternative explanations were excluded by re-evaluation under matched software, and the training loss is flat over the range, so this is not pixel-space overfitting; what does move monotonically is the weight norm, which shrinks by roughly 30% under decoupled weight decay over the $9\times$ longer schedule. Appendix E gives the checks.

Figure 2b shows the corresponding curve under our objective. The panels differ in clip length as well as objective (16 frames against 32), so they are not a controlled comparison; the controlled one is Table 4. What the figure establishes is the direction of each curve.

4.7 ROBUSTNESS AND INFERENCE COST

Two seeds of the 20k model agree to within 0.002 on the five-dataset means and on five of six datasets of the full grid, ETTm1 being the only visible spread (0.351 against 0.360); no second seed of the 60k model exists, so the headline number carries no error bar. Inference is not single-pass everywhere, since the rule of Section 3.5 averages frame scales on short-horizon cells and rolls out beyond one prediction block. Appendix D gives the per-cell pass counts, and Appendix F the variants that did not improve on the reported system, including a two-seed prediction ensemble.

5 DISCUSSION

Two factors, separately supported. Two questions are easy to conflate here. Whether video pre-training is a useful starting point is answered by Section 4.4: yes at a 5k-update budget, where both video initialisations lead the image one and all three lead random initialisation by a wide margin. Whether a transfer design is needed to realise it is answered by Sections 4.5 and 4.6: yes, since the same backbone under its native objective gets worse at forecasting as that objective improves. Our experiments support the two separately; none isolates their interaction, so we do not attribute the end-to-end accuracy to the initialisation alone.

Where the initialisation advantage stops. The advantage does not persist. At 5k updates video initialisation leads by 3–5% at every horizon; at 20k the two are level (Table 3). Video pretraining therefore buys adaptation efficiency rather than a higher ceiling within the budgets we can run, which is why the headline claim is about what a limited pretraining budget reaches and not about video backbones being better in general. The comparison also answers a question about initialisation only: the image-initialised arm is not a stand-in for VisionTS++, which differs in rendering, head and schedule.

Why the forecast-aligned objective helps more at long horizons. Removing the value-space term costs 5% at $H=96$ and 8% at $H=720$ (Table 4). A plausible reading is that short horizons are largely fixed by the visible context, so both criteria rank completions similarly, while longer ones are generated rather than interpolated and a completion can be smooth and pixel-cheap while sitting at the wrong level. The ablation is consistent with this and does not establish it.

Limitations. Mid-to-long horizons on the hourly ETT sets remain behind the strongest baselines: the context is capped at 32 frames of at most four periods, so covering $H=720$ with $P=24$ trades resolution for coverage. Mean MAE is 0.328 against 0.326, which is worse, not equal, and we have no uncertainty estimate that would justify calling it a tie. The evaluation covers one six-dataset protocol; GIFT-Eval (Aksu et al., 2024) and Monash (Godahewa et al., 2021) are not run. Seed replication exists for the 20k model only, so the headline number carries no error bar; budget is reported in corpus size, updates and sample presentations, not FLOPs or wall-clock, and excludes the inherited video pretraining; and the ViT-B backbone comparison is size-matched rather than architecturally exact. Finally, the corpus contains Monash `traffic_hourly`, the same data as the LSF traffic benchmark, so traffic is excluded from our evaluation.

6 CONCLUSION

A backbone pretrained on video is a useful starting point for forecasting but not a usable one: continuing its own objective on rendered series improves that objective while degrading forecasts. What makes the transfer work is a framework built around the forecasting task — period-aligned frames, prediction of a masked future block, a differentiable read-out that puts the loss in value space, and a training scale distribution matched to the inference rule. With it, one checkpoint and one rule reach the lowest mean MSE among the models we compare on the six-dataset zero-shot protocol, after 60k updates on 654M observations. Two results bound that claim: the video initialisation’s lead closes by 20k updates, so what we show is adaptation efficiency rather than a higher ceiling, and the transfer design is what makes any initialisation usable at all. Next steps: a second benchmark, a longer effective context for the ETT horizons we lose, and the same read-out on an image backbone.

REPRODUCIBILITY STATEMENT

Every number in this paper comes from one publicly documented pipeline. The renderer, the value-space read-out, the trainer and all command-line flags used for each arm are described in Section 3 and Appendix A; the evaluation follows the protocol of Chen et al. (2025) and was checked against their released checkpoint to three decimals (Section 4.1). One checkpoint and one inference rule produce all 24 evaluation cells: no per-dataset tuning, no checkpoint ensembling, no blending with statistical baselines. The pretraining corpus is built from public LOTSA subsets (Woo et al., 2024) with a documented subset list, and none of the six evaluation datasets appears in it. Code, corpus manifest and checkpoints will be released.

ETHICS STATEMENT

The work uses public forecasting benchmarks and a public pretraining corpus; no human subjects, no personally identifying data. Electricity and traffic-like series describe aggregate consumption at substation level. The main risk we see is the usual one for forecasting models sold as general purpose: the reported zero-shot accuracy is measured on six datasets in two domains and should not be read as a guarantee on safety-critical deployments.

USE OF LARGE LANGUAGE MODELS

Large language models were used for copy-editing and for scripting routine experiment orchestration. All experimental design, analysis and claims are the authors’; every number reported here was produced by the pipeline described above.

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A IMPLEMENTATION DETAILS

Rendering. Frames are 224×224 grayscale, replicated to three channels. Fill level 0.12, background 0.92, grid lines every 28 rows at 0.80. Values are standardised with the statistics of the visible context, clipped to $z \in [-3, 3]$ and mapped to height by Equation 1. A clip holds $16kP$ observations in F frames; tubelet size is 2, giving $16 \times 14 \times 14 = 3136$ tokens at $F = 32$.

Training. VideoMAE-B initialised from Kinetics-400, `norm_pix_loss` off, AdamW ($\beta_1=0.9$, $\beta_2=0.95$), learning rate 10^{-4} with cosine decay and 500 warm-up steps, weight decay 0.05 (or 0 with prior anchoring at $\lambda_{sp} = 0.05$), bf16, micro-batch 8 with 4-step gradient accumulation for an effective batch of 32. Masked future length h_p is sampled from $\{2, 4\}$; with probability 0.3 the leading tubelets are additionally hidden; 30% of batches use tube masking instead. Frame scale $k \in \{1, 2, 4\}$ with probability 0.5/0.3/0.2. Learning rate 2×10^{-4} is worse and 5×10^{-5} ties 10^{-4} .

Table 5: Our model, per horizon; same checkpoint and inference rule as Table 1. Six-dataset means in the last row.

Dataset	$H=96$		$H=192$		$H=336$		$H=720$		Avg	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.292	0.332	0.332	0.357	0.363	0.377	0.423	0.409	0.352	0.369
ETTh2	0.160	0.245	0.214	0.283	0.264	0.318	0.328	0.362	0.241	0.302
ETTh1	0.352	0.380	0.396	0.408	0.424	0.425	0.446	0.449	0.404	0.416
ETTh2	0.266	0.324	0.333	0.369	0.361	0.395	0.379	0.415	0.335	0.376
ECL	0.128	0.219	0.148	0.239	0.170	0.262	0.207	0.294	0.163	0.254
Weather	0.139	0.183	0.181	0.226	0.232	0.266	0.308	0.331	0.215	0.252
Mean	0.223	0.281	0.267	0.314	0.302	0.341	0.349	0.377	0.285	0.328

Corpus. 98 LOTSA subsets, 184,013 series, 654.2M observations, with 20% synthetic series (sinusoids with trend, noise and regime changes) mixed in. Subsets are sampled with probability proportional to the square root of their observation count. We verified that none of ETTh1, ETTh2, ETTm1, ETTm2, electricity or weather is present; the nearest same-domain entries are Monash weather (daily Australian stations, not the Jena 10-minute set) and `traffic_hourly`, which is why traffic is excluded from the evaluation.

Evaluation. `--mode auto` as described in Section 3.5. All 24 cells use the same checkpoint. Multi-channel datasets are forecast channel by channel. Reported MSE and MAE are computed on standardised values over stride-1 test origins.

B PER-CELL COMPARISON AGAINST VISIONTS

Table 6: MSE per dataset and horizon against VisionTS, the only baseline that publishes per-horizon numbers on this protocol. **Bold:** better of the two. We win 18 of 24 cells; the six losses are all mid-to-long horizons on the hourly ETT sets.

Dataset	$H=96$		$H=192$		$H=336$		$H=720$	
	Ours	VisionTS	Ours	VisionTS	Ours	VisionTS	Ours	VisionTS
ETTh1	0.292	0.341	0.332	0.360	0.363	0.377	0.423	0.416
ETTh2	0.160	0.228	0.214	0.262	0.264	0.293	0.328	0.343
ETTh1	0.352	0.353	0.396	0.392	0.424	0.407	0.446	0.406
ETTh2	0.266	0.271	0.333	0.328	0.361	0.345	0.379	0.388
ECL	0.128	0.177	0.148	0.188	0.170	0.207	0.207	0.256
Weather	0.139	0.220	0.181	0.244	0.232	0.280	0.308	0.330

C INITIALISATION AT TWO CLIP LENGTHS

Table 7: Video against image initialisation at two clip lengths, two seeds each where available. Five-dataset mean MSE at $H=96/336/720$.

Clip length	Video init		Image init	
16 frames	0.257/0.350/0.383,	0.256/0.348/0.384	0.268/0.359/0.400,	0.265/0.353/0.400
32 frames	0.245/0.330/0.379		0.244/0.329/0.373,	0.245/0.330/0.373

At 16 frames the video prior is worth 3–4%; at 32 frames the two initialisations tie. Read together with the anchoring ablation (inert at 32 frames, essential at 16), this suggests that part of what video

pretraining supplies is long-range temporal context, which a longer window supplies directly. The matched-size comparison in Table 3 therefore uses the 5k-step budget, where the prior is still doing work.

D INFERENCE CONFIGURATION AND COST

The rule of Section 3.5 maps (P, H) to a configuration with no per-dataset tuning. Table 8 lists the configuration it selects for each evaluation cell and the number of forward passes that configuration costs. `mp` (multi-period) predicts one block of frames with k periods per frame; the `sc` suffix lists the frame scales that are averaged, each of which is a separate pass; `rollout2` feeds the prediction back and predicts twice. The system uses one checkpoint and one rule for all cells, but it is not single-pass everywhere.

Table 8: Inference configuration and forward passes per evaluation cell.

Dataset	P	$H=96$	$H=192$	$H=336$	$H=720$
ETTh1 / ETTh2	24	<code>sc124</code> (3)	<code>sc12</code> (2)	<code>mp</code> (1)	<code>rollout2</code> (2)
ETTm1 / ETTm2	96	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)
ECL	24	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)
Weather	144	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)	<code>mp</code> (1)

A 32-frame clip costs roughly $8 \times$ a 16-frame clip per pass on the largest dataset, so the inference cost of the reported system is higher than that of a 16-frame configuration even where the pass count is one.

E CHECKS ON THE OBJECTIVE-MISMATCH RESULT

The degradation in Figure 2a survives the following checks. Checkpoint key sets, the `transformers` version, the attention configuration and dataloader warnings were excluded by re-evaluating every checkpoint under matched software. The training loss is flat across the range $(0.0092 \rightarrow 0.0089)$, so the model is not overfitting the pixel objective. Per cell, the 176k-step model is 5–60% worse than the 20k-step model, with the largest gaps at the longest horizons (ETTh2 at $H=720$: $0.429 \rightarrow 0.617$). The one quantity that moves monotonically with the schedule is the weight norm, which shrinks by roughly 30% over the $9 \times$ longer run under decoupled weight decay; this is consistent with the prior-anchoring ablation helping in the short-clip regime, though we did not isolate it as the cause. The 60k native-objective checkpoint was never evaluated downstream, so no value is reported for it.

F ALTERNATIVES THAT DID NOT WORK

We list these so that the obvious questions have answers, not because any of them is a finding.

Prediction-space averaging of two seeds is the one combination that helps, by 0.6% and never worse on any cell; it appears as a secondary row in Table 1.

Table 9: Attempts that did not improve on the recipe in Section 3.

Attempt	Outcome
9× budget under the pixel objective (176k steps)	5–60% worse per cell
Larger batch (128, 256 via accumulation)	equal or worse at every horizon
4× more in-domain data (buildings.900k, LargeST, Q-Traffic, Azure)	ties the corpus we use
Longer masked future (8 of 16 frames)	slightly worse
Value-loss weight 0.3 instead of 0.1	worse
Three-checkpoint ensemble	no gain
Blending with a zero-parameter statistical prior	validation selection does not transfer
Weight-space averaging of two seeds	better on ETT, 16% worse on electricity, net worse
Cross-corpus weight averaging	worse than either parent